

Rule-Grounded Direct-Policy Learning for Pokémon TCG AI

A public-rule semantic projection for factorized legal-action learning

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Claim boundary. This is a compact, submission-ready manuscript template for the design-frozen rule-derivative system. Every numerical statement is classified by evidence type. The iteration-3 formal holdout was waived and measured zero games; it is therefore not described as a pass. The document has not been submitted to Kaggle.

Summary

Competitive Pokémon TCG play is a partially observed control problem whose legal choices change from one prompt to the next. A single action may require an ordered sequence of card, count, source, and target selections. If those options are represented mainly by transient menu indices, two choices with different rule effects can look identical to a learned policy.

We present a direct-policy Alakazam derivative, governed by the revision-27 typed contract [4], that adds a versioned, acting-player-visible rule-semantic projection to an existing causal policy. The pinned competition simulator remains the only authority for legal options, state transitions, and terminal outcomes; the model only ranks the options it is given. The packaged policy uses no RTP, MCTS, or online search. Our central result is therefore about representation and auditability: a corpus census found 127,641 legacy collision groups, while the repaired key left zero unresolved actionable semantic collisions. Gameplay measurements are reported separately and do not establish that this repair alone caused a strength gain.

Why the problem is hard

The controller must preserve three boundaries at once. First, the complete legal option set must survive factorization: a variable-length multi-select cannot be replaced by a small hand-written action vocabulary. Second, only the acting player's information set may affect a decision. Opponent hand identities, deck order, unrevealed Prize identities, and future transitions are forbidden runtime inputs. Third, exact rules stay with the simulator. A learned legality or rule predictor may diagnose errors, but may never create or remove an option.

Design contribution

Our change is deliberately narrow. We preserve the existing card embedding and policy path, then add a zero-initialized residual that encodes exact numeric values, selection context and effect, minimum and maximum counts, remaining resource budgets, stable source/target/area/slot bindings, and pinned public catalog metadata. Unsupported evidence yields a zero residual. With the layer off, the baseline path is reproduced; with it on, distinctions already exposed by the simulator are available to the option scorer. This is neither a card-text parser nor a second game engine.

Data, leakage controls, and model

The sealed corpus covers 20 UTC days from 23 July through 11 August 2026. A row is eligible only when its acting seat's recorded setup deck contains literal card ID 743; the opponent's decisions are not borrowed when only one seat qualifies. The resulting 24,335 games contain 2,351,208 decision occurrences, 18,412,973 option records, and 68 acting-deck variants. Whole days are assigned 14/3/3 to train, validation, and evaluation. The recorded manifests report no cross-split source, day, game, or group overlap.

At inference, a causal Transformer state/history encoder [5] is paired with structured legal option features. Multi-stage selections are decoded one remaining option at a time with an explicit STOP when legal. The public-rule residual is additive to the preserved Card2Vec path, so the repair does not overwrite learned card identity. Eight turn-checklist questions provide factual provenance for hand threshold, safe spending, replacement attackers, forced draws, bench liability, immediate disruption, unknown-Prize robustness, and deck-out timing. At this snapshot, the census accepted only the public-rule semantic projection; the entire checklist-provenance gate branch remains exact-zero and inert. The bench and immediate-disruption channels are additionally contractually trace-only until their narrow predicates pass independent calibration.

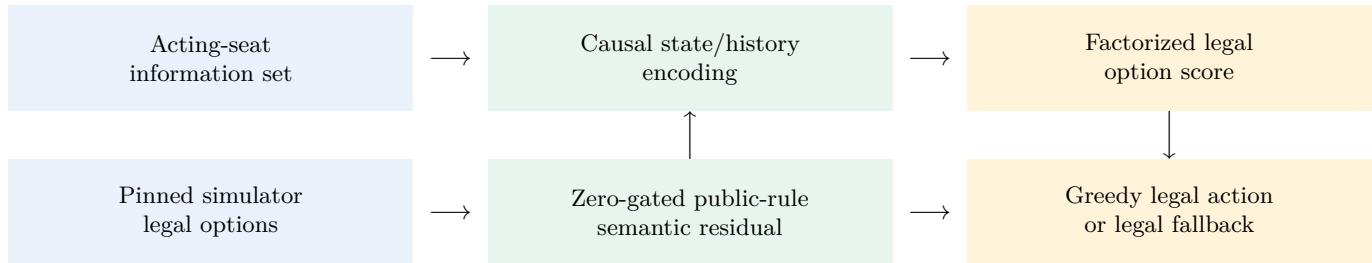


Figure 1: The deployed boundary. Training-only guides and Prize-plan critics do not supply runtime action logits.

Training and evaluation

Immutable r195-compatible weights were mapped into the r274 head structure. The current bug-fixed model then trained for 25 genuine full-corpus epochs: all architecture-present non-combo weights were optimizer-eligible, while combo tensors stayed as ABI-compatible bytes with zero loss and a disabled route. The run visited 57.88 million decision occurrences in 5,757 seconds and ended at semantic loss 1.7141 and semantic accuracy 42.27%. These are optimization diagnostics, not gameplay accuracy.

The post-bootstrap schedule uses an Advantage-Weighted Regression lineage [3] across 20 direct-policy iterations. Each iteration collects 1,024 self-play and 7,172 public-mix games; five full expert refresh epochs occur at iterations 5, 10, 15, and 20, with submissions only at 10 and 20. A separate Prize-plan v2 sidecar predicts public Prize potential at multiple horizons. Its H3 term can shape the selected recorded action during training, but it is frozen, chosen-action-only, and absent from serving.

Verification follows the competition’s simulator boundary and reporting context [2, 1]. It combines simulator witnesses, hidden-information metamorphic pairs, layer-off equality, package smoke tests, collision census, and fixed evaluation identities. A BO250 comparison means 125 seed-matched, seat-swapped pairs and is descriptive until the stricter formal gate is complete. Formal holdouts use 50 games per opponent, split 25/25 by candidate seat, and remain isolated from training and tuning.

Results and evidence status

The collision census is the cleanest design result. The legacy projection aliased 255,398 option records in 127,641 groups; 129,503 selected-action records occurred on those rows. Six semantic-delta classes were observed. After repair, all 99 remaining duplicate groups were harmless permutations and the actionable unresolved count was zero.

One checksum-bound, no-search package (submission 55487412) completed with a public score of 600.0. This single leaderboard observation is not a paired baseline comparison. At iteration 2, a 900-game, 18-opponent holdout recorded a 76.83% skill-weighted win rate and a 74.39% lower bound, but the broader pipeline gate failed its rating-simulation and historical-nonregression checks. It is therefore a measured non-promotion result. Iteration 3’s holdout was explicitly waived: zero formal-holdout games were measured, and no pass is claimed.

Table 1: Evidence is limited to what each artifact supports.

Status	Observation	Supported conclusion
Measured	127,641 legacy groups; 0 actionable repaired groups	Semantic representation repair on the sealed census
Receipt-verified	25 epochs; 42.27% semantic accuracy	Completed full-model optimization, not game strength
Historical Measured	Kaggle 55487412: 600.0 Iteration 2: 900 games, 76.83%; pipeline gate failed	One contextual public score; no causal delta Held-out non-promotion evidence
Not measured	Iteration 3: 0 formal-holdout games	Waiver only; no measured pass

Deck strategy: exact damage without reckless draw

Powerful Hand places two damage counters per card in hand. For a visible target with remaining HP H , the basic knockout threshold is

$$h_{\text{KO}} = \left\lceil \frac{H}{20} \right\rceil,$$

adjusted only for visible effects. The objective is not the largest possible hand; it is the smallest safe hand that secures the prize exchange while preserving a replacement Alakazam line, recovery, bench space, and unavoidable future draws. The default board develops two attacker routes and a Dunsparce line while keeping one bench slot flexible. Deterministic search normally precedes broad draw. Boss’s Orders is saved for a prize-changing or tempo-winning target, and Enhanced Hammer is spent only when a visible Special Energy blocks damage or changes the race. Unknown Prize identities are never guessed. Against a visible 330-HP Mega Lopunny ex, for example, the full-health threshold is 17 cards and the target gives up three Prizes; a visible Mist Energy must be removed or gisted around because it prevents the attack effect. Near deck-out, every mandatory turn, attachment, evolution, and draw effect is counted before the attack is committed.

Limits and reproducibility

This study does not prove leaderboard superiority, causal benefit from the semantic adapter, or strength from the H3 training canary. The current evidence is deck-specific; a public score mixes model, deck, opponent pool, and platform variance. Trace-only checklist channels and shadow RTP artifacts are not described as deployed decision routes.

Every promoted claim is bound to a corpus, schema, checkpoint, package, simulator build, deck multiset, seed/seat schedule, and immutable receipt. The ground-truth engine is Kaggle Environments 1.32.6 with the pinned Linux `libcgl.so`. Source/day/group split manifests, package member manifests, and SHA-256 identities make failed and superseded runs distinguishable from the active derivative. The source archive contains prose, native LaTeX diagrams, and tables only; it does not redistribute the engine, model weights, replays, card art, or external guide text.

Conclusion

Reliable play begins with representing every legal choice distinctly while respecting what the acting player can know. The finalized design combines a pinned simulator, causal observations, factorized legal decoding, a bypassable rule-semantic residual, and receipt-backed evaluation. Its strongest current claim is representation correctness; gameplay gains remain subject to paired, completed evidence.

References

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- [2] Kaggle. Pokémon tcg ai battle: Simulation overview. Competition overview, 2026. Accessed 14 August 2026.
- [3] X. B. Peng, A. Kumar, G. Zhang, and S. Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. *arXiv preprint arXiv:1910.00177*, 2019.
- [4] Project Team. Alakazam elmo rule-derivative contract, revision 27, 2026. Internal typed design authority; SHA-256 68c4e6578db5fe7c1b4d24fe7138f81705a17c8d136e47d7239198a5087c4eba.
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